

Wasserstein distance

JKO scheme: a time-discretized version

From $\mathbb{R}^d$ to $\mathcal{P}(\mathbb{R}^d)$

- Recall implicit Euler scheme:

$$x_{t+1} = \arg \min_x \left\{ f(x) + \frac{1}{2\tau} \|x - x_t\|^2 \right\}$$

- Limit as $\tau \rightarrow 0$: continuous gradient flow in $\mathbb{R}^d$.
- Let's do the same in $\mathcal{P}(\mathbb{R}^d)$! First step: choose a meaningful distance
- Good choice of distance: **Wasserstein distance** W_2
- Leads to the **JKO scheme: a time-discretized version** of the Wasserstein gradient flow in $\mathcal{P}(\mathbb{R}^d)$

JKO Scheme